

$$\text{Q6} \quad \frac{\frac{1}{2}\sqrt{2}}{2dn} = \frac{2dn}{\sqrt{1-4n^2}}$$

$$\sqrt{a^2-x^2} = a \sin \theta$$

$$\sqrt{a^2+x^2} = a \tan \theta$$

$$2n = \sin \theta$$

$$n = \frac{1}{2} \sin \theta$$

$$dn = \frac{1}{2} \cos \theta d\theta$$

$$\frac{\frac{1}{2}\sqrt{2}}{2 \times \frac{1}{4} \cos \theta d\theta} = \frac{2 \times \frac{1}{4} \cos \theta d\theta}{\sqrt{1-4\left(\frac{1}{4} \sin \theta\right)^2}}$$

$$\frac{\frac{1}{2}\sqrt{2}}{\frac{1}{2} \cos \theta d\theta} = \frac{\frac{1}{2} \cos \theta d\theta}{\sqrt{1-4 \times \left(\frac{1}{4}\right)^2 \sin^2 \theta}}$$

$$\frac{\frac{1}{2}\sqrt{2}}{\frac{1}{2} \cos \theta d\theta} = \frac{\frac{1}{2} \cos \theta d\theta}{\sqrt{1-4\left(\frac{1}{16}\right) \sin^2 \theta}}$$

$$\frac{\frac{1}{2}\sqrt{2}}{\frac{1}{2} \cos \theta d\theta} = \frac{\frac{1}{2} \cos \theta d\theta}{\sqrt{1-\frac{1}{4} \sin^2 \theta}}$$

$$\frac{\frac{1}{2}\sqrt{2}}{\frac{1}{2} \cos \theta d\theta} = \frac{\frac{1}{2} \cos \theta d\theta}{\sqrt{\frac{4-1}{4} \sin^2 \theta}}$$

$$\frac{\sqrt{2}}{\frac{1}{2} \cos \theta} = \frac{\frac{1}{2} \cos \theta}{\sqrt{\frac{3}{4} \sin^2 \theta}}$$

$$\frac{\frac{1}{2}\sqrt{2}}{2 \times \frac{1}{2} \cos \theta d\theta} = \frac{2 \times \frac{1}{2} \cos \theta d\theta}{\sqrt{1-4\left(\frac{1}{2} \sin \theta\right)^2}}$$

$$= \frac{\cos \theta d\theta}{\sqrt{1-4 \times \frac{1}{4} \sin^2 \theta}}$$

$$= \frac{\cos \theta d\theta}{\sqrt{1-\sin^2 \theta}}$$

$$= \frac{\cos \theta d\theta}{\sqrt{\cos^2 \theta}}$$

$$= \frac{\cos \theta d\theta}{\cos \theta}$$

$$= \int_0^{\frac{\pi}{4}} d\theta = \left[ \theta \right]_0^{\frac{\pi}{4}}$$

$$= \left( \frac{\pi}{4} \right) - (0) = \frac{\pi}{4}$$

Ans.